

Solution: Let f be strictly decreasing. By Defn. 6, $\forall x \forall y, x, y \in \mathbb{R}$ if $x < y$, then $f(x) > f(y)$. We know that $f(x)f(y) \neq 0$.

$1/f(x) < 1/f(y) \Rightarrow g(x) < g(y)$. By Defn. 6, g is strictly increasing.

Let g be strictly increasing. By Defn. 6, $\forall x \forall y, x, y \in \mathbb{R}$, if $x < y$, then $g(x) < g(y)$.

$\therefore 1/f(x) < 1/f(y) \Rightarrow f(x) > f(y)$. By Defn. 6, f is strictly decreasing.

□

26) Prove that a strictly increasing function from $\mathbb{R}$ to itself is one-to-one.

Solution: Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a strictly increasing function. By Defn. 6, $\forall x \forall y, x, y \in \mathbb{R}$ if $x < y$, then $f(x) < f(y)$.

Let $\exists a, b \in \mathbb{R}$ s.t. $f(a) = f(b)$. If $a > b$, then by Defn. 6, $f(a) > f(b)$.

If $a < b$, then by Defn. 6, $f(a) < f(b)$.

$\therefore a = b$. By Defn. 5, f is one-to-one. □

b) Give an example of an increasing function from $\mathbb{R}$ to itself that is not one-to-one.

Solution: $f(x) = \lfloor x \rfloor$. We shall show that this is an increasing function $f: \mathbb{R} \rightarrow \mathbb{R}$. Let $x, y \in \mathbb{R}$. WLOG, let $x < y$.

$f(x) = \lfloor x \rfloor$. Let $\lfloor x \rfloor = n$. $\therefore x-1 < n \leq x$ (By Property 1(c) from Table 1)

$f(y) = \lfloor y \rfloor$. Let $\lfloor y \rfloor = m$. $\therefore y-1 < m \leq y$ (By Property 1(c) from Table 1)

$\therefore x < y \therefore x-1 < y-1$ By Property 1(a) from Table 1, we have:
 $n \leq x < n+1$ and $m \leq y < m+1$

$\therefore \because x < y, n \leq m. \Rightarrow \lfloor x \rfloor \leq \lfloor y \rfloor. \Rightarrow f(x) \leq f(y)$

$\therefore f(x) = \lfloor x \rfloor$ is an increasing function. (By Defn. 6)

We know that $1.1, 1.2 \in \mathbb{R}$. $f(1.1) = \lfloor 1.1 \rfloor = 1$ $f(1.2) = \lfloor 1.2 \rfloor = 1$

But $1.1 \neq 1.2$. From Defn. 5, f is not one-to-one. □

27) Prove that a strictly decreasing function from $\mathbb{R}$ to itself is one-to-one.